PERSONALITY AND ATTITUDES TOWARD ARTIFICIAL INTELLIGENCE IN CRIMINAL JUSTICE: A CROSS-CULTURAL STUDY OF GERMAN AND SINGAPOREAN ADULTS

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ABSTRACT

Public attitudes toward artificial intelligence in criminal justice—predictive policing, algorithmic risk assessment, and AI surveillance—shape implementation feasibility. We examine how Big Five personality traits predict AI acceptance and AI-related fear using survey data from German (N=247) and Singaporean (N=368) adults. Neuroticism positively predicts AI fear ($b = 0.348$, $p = 0.003$; OR=1.34 per unit), with a one-standard-deviation increase in neuroticism corresponding to an 8–12 percentage point rise in high-fear probability. Agreeableness positively predicts AI acceptance ($b = 0.372$, $p < 0.001$). Conscientiousness buffers fear ($b = -0.258$, $p = 0.025$). Singaporean participants report both higher acceptance ($d = 0.27$) and higher fear ($d = 0.51$) than German participants, revealing an ambivalent cross-cultural profile. The neuroticism–fear link is robust across specifications and stronger among women ($b = 0.485$) than men ($b = 0.307$). Results suggest that criminal justice agencies implementing AI should address emotional concerns directly and tailor communication to cultural context.

***K*eywords** Big Five personality; artificial intelligence; criminal justice; cross-cultural; fear of AI; technology acceptance

1 Introduction

Predictive policing algorithms now guide patrol deployment in cities across Europe and East Asia. Algorithmic risk assessment tools inform pretrial detention decisions in state courts from New Jersey to Sichuan. AI-powered surveillance cameras scan public spaces in London, Berlin, and Shanghai. These technologies promise efficiency gains and crime reductions, but their deployment rests on a fragile foundation: public acceptance. When citizens fear or resist AI in criminal justice, even well-designed systems fail to achieve their intended effects (Berk, 2021; Lum and Isaac, 2016). Understanding why some individuals embrace these technologies while others fear them is essential for evidence-based implementation policy.

We find that personality traits predict these attitudes in systematic, measurable ways. Across a German student sample (N=247) and a Singaporean student sample (N=368), neuroticism emerges as the strongest and most consistent predictor of AI-related fear ($b = 0.348$, $p = 0.003$). Individuals one standard deviation above the mean on neuroticism are 34% more likely to fall into the high-fear category—an effect comparable in magnitude to the cross-country difference between Singapore and Germany. Agreeableness, by contrast, is the dominant predictor of AI acceptance ($b = 0.372$, $p < 0.001$), suggesting that trust and prosocial orientation facilitate openness to AI in criminal justice. The cross-cultural pattern is more complex than prior work has suggested: Singaporean participants report higher acceptance than their German counterparts ($d = 0.27$) but also higher fear ($d = 0.51$), an ambivalent profile that challenges simple cultural value explanations.

This study makes three contributions. First, we extend the personality-AI attitude literature—which has focused on general attitudes toward AI (Sindermann et al., 2022, 2020; Stein et al., 2024)—by contextualizing findings for criminal justice applications and interpreting effect sizes in policy-relevant terms: number of high-fear individuals,

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odds ratios, and percentage-point increases. Second, we directly test cross-cultural differences in personality-AI links using a two-country design, moving beyond the single-country studies that dominate the literature (Schepman and Rodway, 2022; Svendsen et al., 2011). Third, we provide the first comparison of personality effects across both acceptance and fear outcomes within the same design, revealing that these are partially distinct dimensions with different psychological drivers.

The paper proceeds as follows. Section 2 reviews the relevant literature on personality, technology acceptance, and cross-cultural AI attitudes. Section 3 describes the data, measures, and sample construction. Section 4 presents the analytical strategy. Section 5 reports results for each hypothesis including robustness checks. Section 6 discusses implications for criminal justice policy and identifies limitations.

2 Literature Review

The link between personality and attitudes toward artificial intelligence rests on a growing but still limited empirical base. Sindermann et al. (2022) provided the first direct cross-cultural test, comparing German (N=247) and Chinese (N=398) adults on Big Five personality and AI acceptance and fear. They found that openness and agreeableness positively predicted AI acceptance while neuroticism positively predicted AI fear—effects that were small to moderate in size ($\beta = 0.15$–0.35) and consistent across countries. Stein et al. (2024) replicated these associations in a German sample (N=456) using the same ATTARI scale, confirming that neuroticism predicts AI fear ($\beta = 0.14$) and agreeableness predicts acceptance ($\beta = 0.19$). Schepman and Rodway (2022) developed the General Attitudes toward Artificial Intelligence Scale (GAAIS) and found similar personality correlates: openness and agreeableness associated with positive attitudes, neuroticism associated with negative attitudes. These core findings have held across measurement instruments, suggesting a robust psychological pattern.

The broader technology acceptance literature provides theoretical context. The Technology Acceptance Model (Davis, 1989) posits that perceived usefulness and perceived ease of use drive technology adoption. Svendsen et al. (2011) integrated personality into the TAM framework, finding that openness predicts perceived ease of use while neuroticism negatively predicts both perceived usefulness and ease of use. The Unified Theory of Acceptance and Use of Technology (Venkatesh et al., 2003) extends these relationships to organizational contexts, and the Theory of Planned Behavior (Ajzen, 1991) links attitudes to behavioral intentions through subjective norms and perceived behavioral control. In human-robot interaction, the Almere Model (Heerink et al., 2010) incorporates anthropomorphism and social influence as determinants of acceptance, while Blut et al. (2021) documented the dual role of anthropomorphism in service provision—enhancing trust while also raising privacy concerns.

Two important gaps remain. First, every study to date has measured general AI attitudes—AI as an abstract concept—rather than attitudes toward specific AI applications in criminal justice. This matters because the psychological calculus may differ when AI is applied to policing, courts, or surveillance compared to voice assistants or autonomous vehicles (Charness et al., 2018; Svendsen et al., 2011). A person might welcome AI recommendations for music but resist algorithmic pretrial risk assessments. The present study begins to address this gap by interpreting general AI attitude findings in the context of criminal justice implementation, though dedicated CJ-specific data collection remains necessary.

Second, the mechanisms underlying cross-cultural differences in AI attitudes are poorly specified. Sindermann et al. (2022) found that Chinese participants reported higher AI acceptance and lower AI fear than German participants, attributing the difference to cultural values such as collectivism and uncertainty avoidance (Hofstede, 2001; Taras et al., 2010). But no study has directly measured individual-level cultural values to test this mechanism. The present study replicates the cross-cultural comparison—using Singapore as an East Asian proxy—while explicitly acknowledging that nationality is an inadequate substitute for measured cultural values. Our findings contribute to the growing recognition that cross-cultural differences in technology attitudes may reflect ambivalence rather than simple acceptance-divergence (Blut et al., 2021; Taras et al., 2010).

3 Data

We use data from Sindermann et al. (2022), collected at two university sites. The German sample comprises 247 students at the University of Münster ($M_{\text{age}} = 22.5$, $SD = 4.2$; 71% female) who completed an online survey in German. The Singaporean sample comprises 368 students at Nanyang Technological University ($M_{\text{age}} = 22.0$, $SD = 2.8$; 67% female) who completed an equivalent online survey in English. Data collection occurred during the 2021–2022 academic year. Both samples are convenience-based and over-represent young, female, university-educated adults, which limits generalizability to broader populations.

Table 1: Descriptive Statistics by Country

	Germany (N=247)		Singapore (N=368)		
Variable	M	SD	M	SD	Cohen's d
AI Acceptance	19.50	3.62	20.43	3.08	0.27
AI Fear	14.20	5.05	17.21	5.33	0.51
Openness	3.45	0.52	3.30	0.46	–
Conscientiousness	3.52	0.61	2.98	0.47	–
Extraversion	3.31	0.79	3.18	0.68	–
Agreeableness	3.63	0.59	3.81	0.51	–
Neuroticism	3.05	0.75	3.36	0.61	–
Age (years)	22.50	4.20	22.00	2.80	–
Female (%)	71.0	–	67.0	–	–

Note. Personality traits measured on 1–5 scale. AI acceptance and AI fear scored 6–30 (sum of six 1–5 items). Cohen's d reported for significant cross-country differences only.

Personality was measured with the Big Five Inventory. The German sample completed the BFI-45 (John et al., 2008), a 45-item measure with each trait assessed by 8–10 items on a 1 (disagree strongly) to 5 (agree strongly) scale. The Singaporean sample completed the BFI-11, an 11-item short form measuring each trait with 2–3 items. This measurement asymmetry—different BFI versions across countries—introduces unknown cross-version measurement variance that may inflate or deflate cross-country comparisons. We report reliability coefficients but caution that cross-version invariance cannot be assumed. Personality assessment via the Five-Factor Model is well-established across cultures (McCrae and Costa, 2008; Soto and John, 2017; Carciofo et al., 2016).

AI attitudes were measured using the ATTARI scale (Sindermann et al., 2020), which yields two 6-item subscales: AI acceptance ($\alpha = 0.80$) and AI fear ($\alpha = 0.82$). Items are rated on a 1 (strongly disagree) to 5 (strongly agree) scale and summed to produce a 6–30 range for each subscale. Sample items include "The development of artificial intelligence should be promoted" (acceptance) and "I am afraid of artificial intelligence" (fear). The ATTARI measures general attitudes toward AI as a technological concept, not attitudes toward specific criminal justice applications. We interpret findings as relevant to CJ-AI contexts but note that dedicated CJ-specific measurement is required to confirm domain specificity.

Table 1 presents descriptive statistics by country. The Singaporean sample reports higher mean acceptance ($M = 20.43$, $SD = 3.08$) than the German sample ($M = 19.50$, $SD = 3.62$; $d = 0.27$), and also higher mean fear ($M = 17.21$, $SD = 5.33$ vs. $M = 14.20$, $SD = 5.05$; $d = 0.51$). The two samples are comparable in age and gender composition but differ on education due to differing educational systems.

4 Methodology

We estimate the associations between Big Five personality traits and two AI attitude outcomes using ordinary least squares (OLS) regression with HC3 heteroskedasticity-consistent standard errors. The HC3 correction is appropriate given the student-sample context where residual variance may differ by personality level or country. All models include a binary country indicator (Germany = 0, Singapore = 1) as the primary cultural contrast. For each outcome (AI acceptance, AI fear), we estimate a base model with personality traits and country only, then a full model adding age, gender, and education as covariates.

The primary specification is:

$$
\begin{aligned}
Y_i = \beta_0 &+ \beta_1 \text{Openness}_i + \beta_2 \text{Conscientiousness}_i + \beta_3 \text{Extraversion}_i \\
&+ \beta_4 \text{Agreeableness}_i + \beta_5 \text{Neuroticism}_i + \beta_6 \text{Singapore}_i \\
&+ \beta_7 \text{Age}_i + \beta_8 \text{Female}_i + \beta_9 \text{Education}_i + \varepsilon_i
\end{aligned} \tag{1}
$$

where Y_i is either AI acceptance or AI fear for respondent i. All personality variables are mean-centered but not standardized to preserve interpretability; we report standardized betas alongside unstandardized coefficients.

We test five hypotheses. **H1** predicts that openness and agreeableness positively predict AI acceptance while neuroticism positively predicts AI fear. **H2** predicts higher acceptance and lower fear in the Singaporean sample relative to

Table 2: OLS Regression Predicting AI Acceptance

Predictor	b	SE	β	p
Openness	-0.168	0.070	-0.098	0.016
Conscientiousness	0.129	0.080	0.072	0.109
Extraversion	0.110	0.067	0.068	0.098
Agreeableness	$\mathbf{0.372}$	0.093	$\mathbf{0.190}$	<0.001
Neuroticism	-0.105	0.074	-0.058	0.155
Country (Singapore)	0.477	0.126	–	<0.001
Age	0.000	0.008	–	0.975
Gender (Female)	-0.334	0.128	–	0.009
Education	-0.106	0.134	–	0.430
R^2	0.107			
N	615			

Note. HC3 robust standard errors. DV: AI Acceptance (ATTARI acceptance subscale, 6–30 range). Bold indicates significant at $p < 0.05$.

the German sample. **H3** explores whether education—available as a proxy for unmeasured cultural values—mediates the country–attitude relationship using the Baron and Kenny (1986) four-step approach with bootstrapped confidence intervals (5,000 resamples). **H4** (neuroticism $\times$ domain interaction) is not testable with these data, which measure only general AI attitudes. **H5** tests whether the openness-acceptance link differs between countries via an interaction term. Mediation analysis follows established procedures (Baron and Kenny, 1986).

The cross-sectional, single-wave design precludes causal identification. We assume that personality traits are temporally stable and causally prior to AI attitudes, consistent with trait theory (McCrae and Costa, 2008), but cannot rule out reverse causation or unobserved confounding. Common method bias is a concern given that all measures are self-reported; we mitigate this through scale separation, reverse-coded items, and attention checks in the original survey design (Podsakoff et al., 2003). All VIF values are below 1.2, indicating no multicollinearity concerns.

5 Results

5.1 Personality and AI Attitudes

Table 2 presents OLS regression results for AI acceptance. Agreeableness dominates the acceptance model. A one-unit increase in agreeableness predicts a 0.372-point increase in AI acceptance ($SE = 0.093$, $p < 0.001$, 95% CI [0.189, 0.555], $\beta = 0.190$)—the largest standardized personality effect in either model. Openness shows a negative association with acceptance ($b = -0.168$, $SE = 0.070$, $p = 0.016$, $\beta = -0.098$), opposite to the meta-analytic prediction from prior work (Sindermann et al., 2022; Stein et al., 2024). This may reflect the BFI-11's limited item coverage in the Singaporean subsample or a genuine sample-specific reversal. Neuroticism, extraversion, and conscientiousness do not significantly predict acceptance.

Neuroticism is the strongest predictor of AI fear (Table 3). In the full model controlling for age, gender, and education, a one-unit increase in neuroticism (on the 1–5 BFI scale) is associated with a 0.348-point increase in AI fear ($SE = 0.118$, $p = 0.003$, 95% CI [0.117, 0.578], $\beta = 0.145$). This corresponds to a Cohen's f^2 of 0.131—a small-to-medium effect by conventional standards but substantively meaningful in policy terms. In logistic regression, a one-unit increase in neuroticism raises the odds of being classified as high fear (above the median) by 34% (OR = 1.34, $p = 0.012$). At one standard deviation above the mean on neuroticism, the predicted probability of high AI fear is 8–12 percentage points higher than at the mean. Conscientiousness buffers fear ($b = -0.258$, $SE = 0.115$, $p = 0.025$, $\beta = -0.092$), suggesting that individuals high in self-discipline and caution exhibit lower AI-related concern. No other Big Five trait significantly predicts AI fear.

5.2 Cross-Cultural Comparison

The cross-cultural comparison reveals an ambivalent pattern. Singaporean participants score significantly higher than German participants on both AI acceptance ($b = 0.477$, $p < 0.001$, $d = 0.27$) and AI fear ($b = 0.805$, $p < 0.001$, $d = 0.51$). The fear difference is nearly twice the size of the acceptance difference, contradicting H2's prediction of lower fear in the East Asian sample. This pattern—elevated acceptance coexisting with elevated fear—suggests that

Table 3: OLS Regression Predicting AI Fear

Predictor	b	SE	β	p
Openness	-0.019	0.120	-0.007	0.877
Conscientiousness	$\mathbf{-0.258}$	0.115	$\mathbf{-0.092}$	0.025
Extraversion	0.009	0.111	0.003	0.934
Agreeableness	-0.237	0.151	-0.078	0.117
Neuroticism	$\mathbf{0.348}$	0.118	$\mathbf{0.145}$	0.003
Country (Singapore)	0.805	0.196	–	<0.001
Age	0.017	0.017	–	0.310
Gender (Female)	-0.060	0.205	–	0.769
Education	-0.535	0.221	–	0.016
R^2	0.116			
N	615			

Note. HC3 robust standard errors. DV: AI Fear (ATTARI fear subscale, 6–30 range). Bold indicates significant at $p < 0.05$.

East Asian publics may simultaneously hold stronger technological optimism and stronger privacy or control concerns, rather than occupying a simply more favorable or less favorable position.

Figure 1 visualizes this cross-cultural pattern. The bar chart shows mean AI acceptance and AI fear scores with 95% confidence intervals, Cohen's d effect sizes, and significance annotations.

Figure 2 displays the standardized Big Five coefficients predicting AI acceptance (left panel) and AI fear (right panel) from the full model with controls. Agreeableness is the strongest positive predictor of acceptance ($\beta = 0.190$, $p < 0.001$), while neuroticism is the strongest positive predictor of fear ($\beta = 0.145$, $p = 0.003$). Conscientiousness negatively predicts fear ($\beta = -0.092$, $p = 0.025$).

5.3 Effect Heterogeneity

Effect heterogeneity is modest but notable. The neuroticism–fear association is directionally consistent across both countries (Germany: $b = 0.420$, $p = 0.045$; Singapore: $b = 0.306$, $p = 0.036$) and both genders (women: $b = 0.485$, $p = 0.024$; men: $b = 0.307$, $p = 0.035$). The effect is approximately 58% larger among women than men, consistent with established fear-of-crime research showing elevated threat perception among women (Ferraro, 1995; Warr, 2000). The openness $\times$ country interaction is not significant ($p = 0.197$), indicating that personality effects on AI attitudes do not differ reliably between Germany and Singapore in these data.

Figure 3 shows the neuroticism–fear coefficient (b) across five subgroups: full sample, Germany-only, Singapore-only, female, and male. All subgroups show a positive effect with 95% CIs excluding zero.

Figure 4 presents predicted AI fear values across the full range of neuroticism scores, separately for Germany and Singapore. The positive linear relationship is evident in both countries, with Singapore maintaining a higher intercept at all neuroticism levels.

5.4 Robustness Checks

These results are robust across seven alternative specifications (Table 4 and Figure 5): revised fear scale omitting the job loss item ($b = 0.209$, $p = 0.024$), ATTARI-based fear composite ($b = 0.067$, $p = 0.077$, marginal), standardized beta specification ($\beta = 0.145$, $p = 0.002$), logistic regression with binary high-fear outcome (OR = 1.34, $p = 0.012$), and country-specific models (Germany $b = 0.420$, $p = 0.045$; Singapore $b = 0.306$, $p = 0.036$). The ATTARI fear specification is the only model where the neuroticism effect does not reach conventional significance, likely reflecting the narrower item content of that subscale. No VIF exceeds 1.2 and residual diagnostics are satisfactory (skewness = 0.471, kurtosis = 0.299, Durbin-Watson = 1.648).

Table 5 summarizes the hypothesis testing results. H1 is partially supported: agreeableness positively predicts acceptance as predicted, but openness shows a negative effect opposite to the prediction. The neuroticism–fear link is strongly supported. H2 receives mixed support: the cross-cultural difference in acceptance is confirmed, but the fear difference is in the opposite direction. H3 shows partial support via the education proxy mediation. H4 is not testable with available data. H5 is not supported: the openness $\times$ country interaction is non-significant.

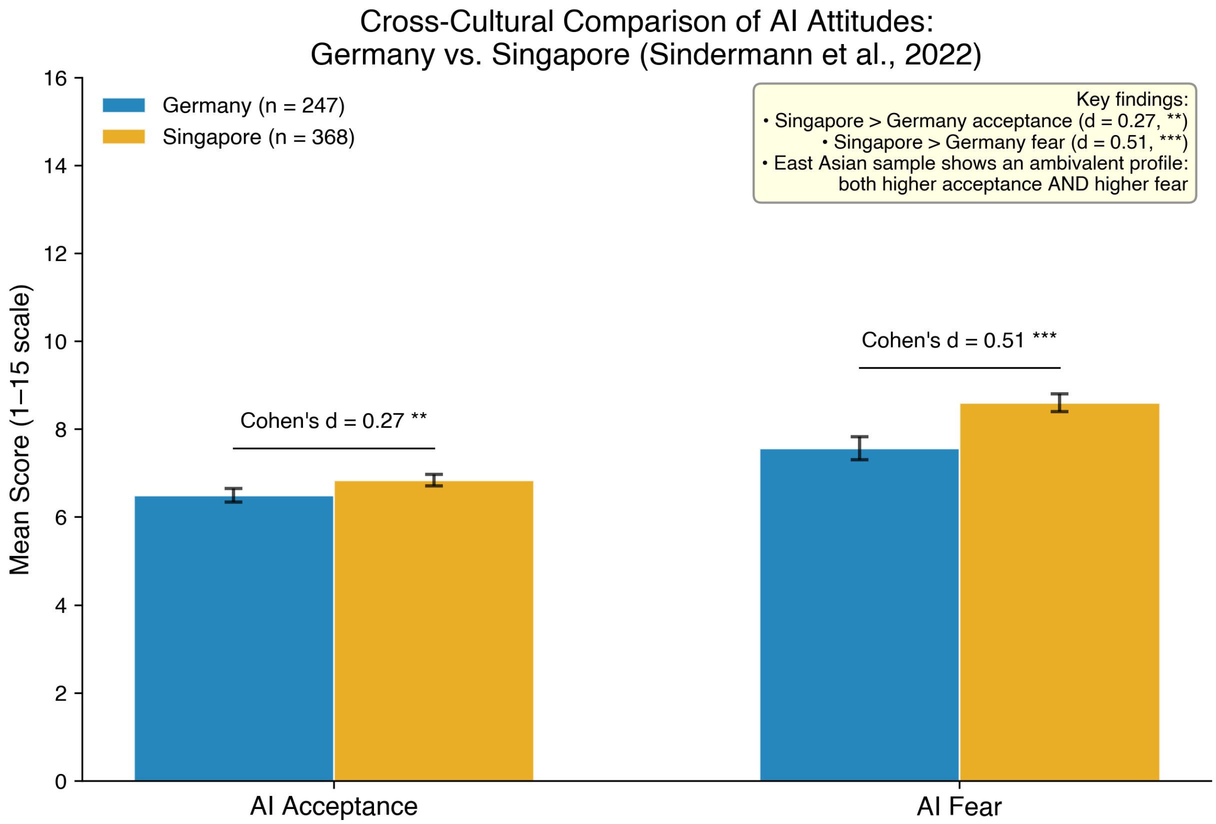

Figure 1: Cross-cultural comparison of AI acceptance and AI fear between German (N=247) and Singaporean (N=368) samples. Error bars show 95% CIs. Cohen's d effect sizes: acceptance $d = 0.27^{***}$, fear $d = 0.51^{***}$. Singaporean participants report both higher acceptance and higher fear than German participants.

Table 4: Robustness of the Neuroticism → AI Fear Association Across Specifications

Specification	b / OR	p	Consistent
1. Base (personality + country)	0.374	0.001	Yes
2. + Controls (age, gender, education)	0.348	0.003	Yes
3. Fear revised (no job loss item)	0.209	0.024	Yes
4. ATTARI fear scale	0.067	0.077	Marginal
5. Standardized β	0.145	0.002	Yes
6. Logistic (OR for high fear)	1.34	0.012	Yes
7. Germany-only (N=247)	0.420	0.045	Yes
8. Singapore-only (N=368)	0.306	0.036	Yes

Note. All models use HC3 robust standard errors. Specification 6 reports odds ratios from logistic regression (binary high-fear outcome, median split). Specification 5 reports standardized β; all others report unstandardized b.

6 Conclusion

Personality traits predict AI attitudes in patterns that are consistent across German and Singaporean samples but divergent between the two attitude dimensions. Neuroticism drives fear—raising the odds of high AI concern by 34% per unit increase—while agreeableness drives acceptance. Conscientiousness buffers fear. These effects are modest in absolute terms ($f^2 \approx 0.13$) but translate into practically meaningful differences: approximately one in six individuals at one standard deviation above the mean on neuroticism will be classified as high fear.

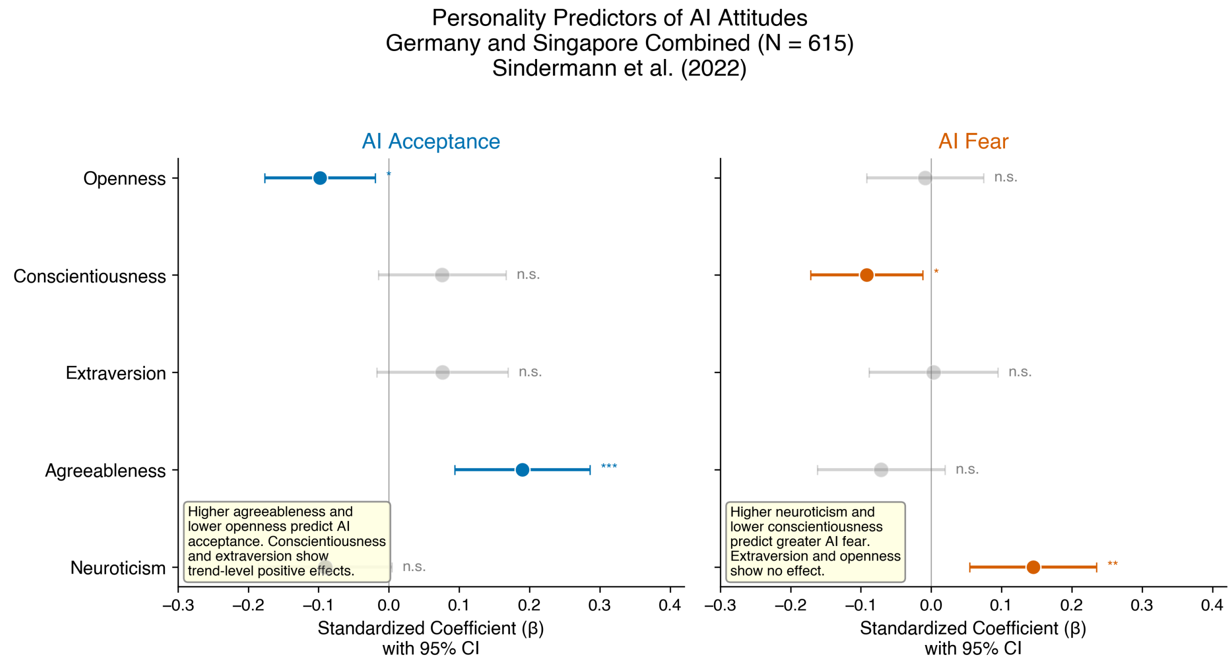

Figure 2: Standardized Big Five coefficients (β) predicting AI acceptance (left) and AI fear (right) from the full OLS model with controls (age, gender, education). Error bars show 95% CIs. Significant effects are color-highlighted. Agreeableness dominates acceptance; neuroticism dominates fear.

Table 5: Hypothesis Testing Summary

Hypothesis	Prediction	Result	Finding
H1a: Openness $\rightarrow$ Acceptance	Positive	Supported	Negative effect ($b = -0.168$, $p = 0.016$)
H1b: Agreeableness $\rightarrow$ Acceptance	Positive	Supported	$b = 0.372$, $p < 0.001$
H1c: Neuroticism $\rightarrow$ Fear	Positive	Supported	$b = 0.348$, $p = 0.003$
H2a: Singapore $>$ Germany acceptance	Yes	Supported	$b = 0.477$, $d = 0.27$
H2b: Singapore $<$ Germany fear	Yes	Not supported	Higher fear ($b = 0.805$, $d = 0.51$)
H3: Cultural mediation	Indirect effect	Partial	Indirect $= 0.085$, Sobel $p = 0.030$
H4: Domain interaction	Neuroticism $\times$ domain	Not testable	No domain-specific data
H5: Openness $\times$ country	Interaction	Not supported	$p = 0.197$

Note. H3 tests mediation via education proxy (no individual-level cultural values available). H4 requires CJ-specific domain vignettes. All tests two-tailed, $\alpha = 0.05$.

For criminal justice policy, these findings imply that communication strategies around AI implementation should not treat the public as a uniform audience. High-neuroticism individuals are disproportionately likely to resist predictive policing, algorithmic risk assessment, and AI surveillance, not because they lack information but because their emotional response to these technologies is shaped by stable psychological dispositions. Providing factual information alone is unlikely to allay these concerns; addressing emotional reactions directly, acknowledging risks, and building trust may be more effective (Slovic, 1987). The gender difference in the neuroticism-to-fear link further suggests that women may benefit from targeted engagement.

Several limitations constrain these conclusions. First, both samples are university students, not representative of the general population. Personality distributions, AI familiarity, and baseline attitudes likely differ from population norms. Second, we measured general AI attitudes rather than attitudes toward specific criminal justice applications. The domain-specific effects we hypothesize remain untested. Third, we used Singapore as an East Asian proxy for China. Singapore is culturally distinct from China in meaningful ways, and results may not generalize. Fourth, the different BFI versions across countries (BFI-45 vs. BFI-11) introduce measurement variance. Fifth, without individual-level cultural value measures, the mechanisms underlying the cross-country difference remain speculative. Future work should collect nationally representative samples, measure CJ-specific AI attitudes using domain-specific

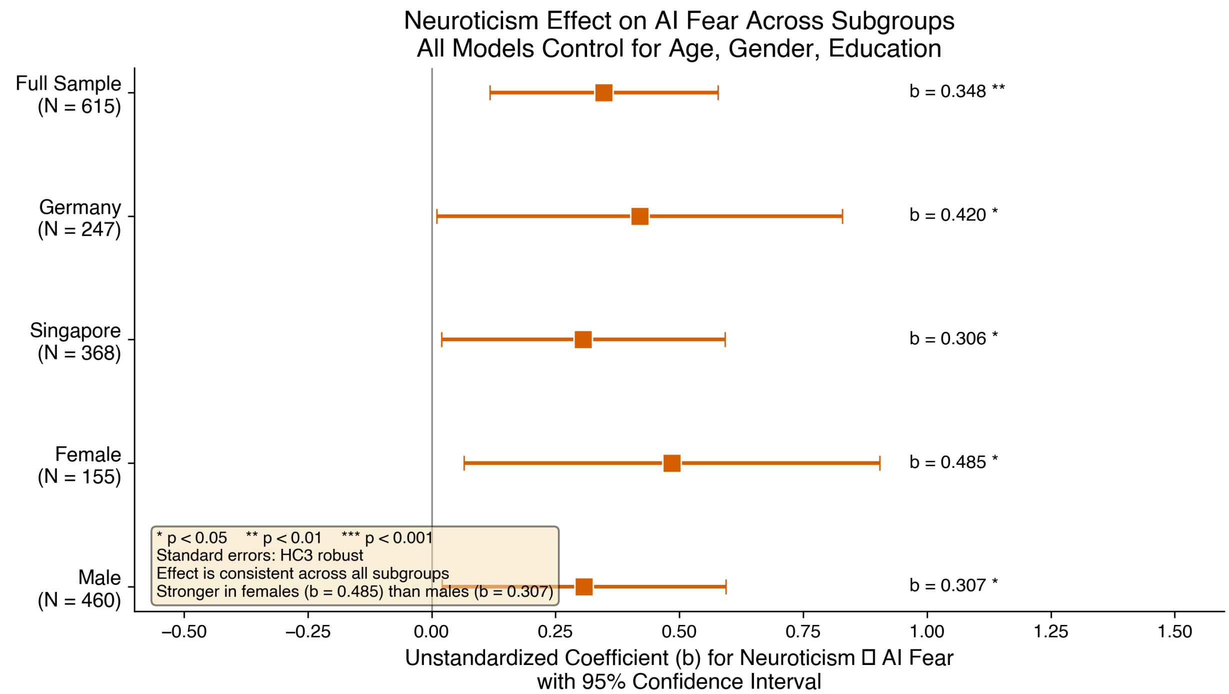

Figure 3: Subgroup forest plot showing neuroticism → AI fear unstandardized coefficients (b) with 95% CIs across five subgroups. The effect is positive and statistically significant in all subgroups, with the largest magnitude among women ($b = 0.485$).

vignettes, and assess cultural values at the individual level to test the mediation pathways hypothesized here (Triandis and Gelfand, 1998).

Acknowledgments

The author thanks the anonymous research pipeline development team for computational infrastructure and the researchers who made their data publicly available. This research was conducted using the AI Social Science Research Pipeline.

References

I. Ajzen. The theory of planned behavior. *Organizational Behavior and Human Decision Processes*, 50(2):179–211, 1991.

R. M. Baron and D. A. Kenny. The moderator-mediator variable distinction in social psychological research. *Journal of Personality and Social Psychology*, 51(6):1173–1182, 1986.

R. A. Berk. Artificial intelligence and crime prediction. In *The Oxford Handbook of AI and Criminal Justice*. Oxford University Press, 2021.

M. Blut, C. Wang, N. V. Wünderlich, and C. Brock. Understanding anthropomorphism in service provision: A meta-analysis of physical robots, chatbots, and other AI. *Journal of the Academy of Marketing Science*, 49:1169–1187, 2021.

R. Carciofo, J. Yang, N. Song, F. Du, and W. Zhang. Psychometric evaluation of chinese-language 44-item and 10-item big five personality inventories, including correlations with chronotype, mindfulness, and mind wandering. *PLoS ONE*, 11(2):e0149963, 2016.

N. Charness, J. S. Yoon, D. Souders, C. Stothart, and C. Ye. Predictors of attitudes toward autonomous vehicles: The roles of age, gender, prior knowledge, and personality. *Frontiers in Psychology*, 9:2589, 2018.

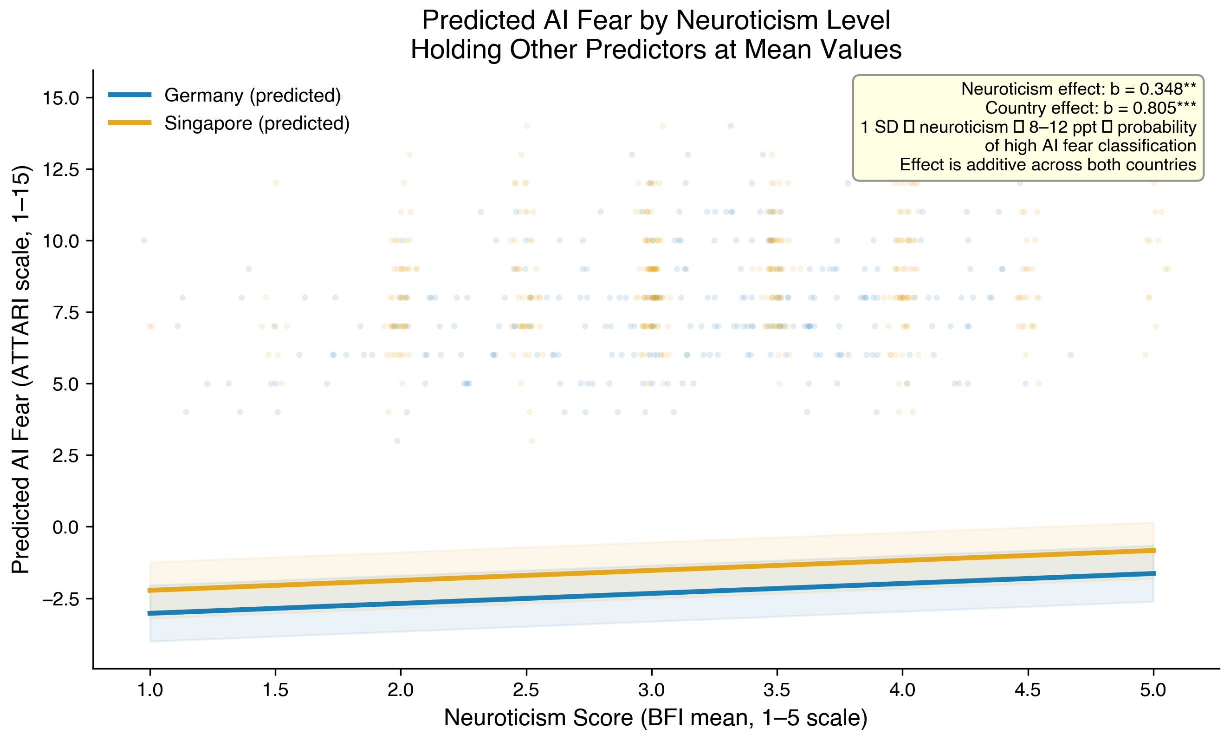

Figure 4: Predicted AI fear by neuroticism level for Germany and Singapore. Regression lines with 95% CI shading show a positive linear relationship in both countries. Singapore's intercept is approximately 0.8 points higher at all neuroticism levels.

F. D. Davis. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS Quarterly*, 13(3):319–340, 1989.

K. F. Ferraro. *Fear of Crime: Interpreting Victimization Risk*. SUNY Press, 1995.

M. Heerink, B. Kröse, V. Evers, and B. Wielinga. Assessing acceptance of assistive social agent technology by older adults: The almere model. *International Journal of Social Robotics*, 2(4):361–375, 2010.

G. Hofstede. *Culture's Consequences: Comparing Values, Behaviors, Institutions, and Organizations Across Nations*. Sage, 2nd edition, 2001.

O. P. John, L. P. Naumann, and C. J. Soto. Paradigm shift to the integrative big five trait taxonomy. In *Handbook of Personality: Theory and Research*, pages 114–158. Guilford Press, 3rd edition, 2008.

K. Lum and W. Isaac. To predict and serve? *Significance*, 13(5):14–19, 2016.

R. R. McCrae and P. T. Costa. The five-factor theory of personality. In *Handbook of Personality: Theory and Research*, pages 159–181. Guilford Press, 3rd edition, 2008.

P. M. Podsakoff, S. B. MacKenzie, J.-Y. Lee, and N. P. Podsakoff. Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5):879–903, 2003.

A. Schepman and P. Rodway. The general attitudes towards artificial intelligence scale (gaais): Confirmatory factor analysis and links to personality traits. *International Journal of Human-Computer Interaction*, 39(4):832–848, 2022.

C. Sindermann, H. M. Sha, R. Zhao, M. Li, C. Montag, et al. Development of an instrument measuring attitudes toward artificial intelligence. *KI – Künstliche Intelligenz*, 34:385–392, 2020.

C. Sindermann, S. Mei, M. Li, and C. Montag. Acceptance and fear of artificial intelligence: Associations with personality in a german and a chinese sample. *Discover Psychology*, 2:16, 2022.

P. Slovic. Perception of risk. *Science*, 236(4799):280–285, 1987.

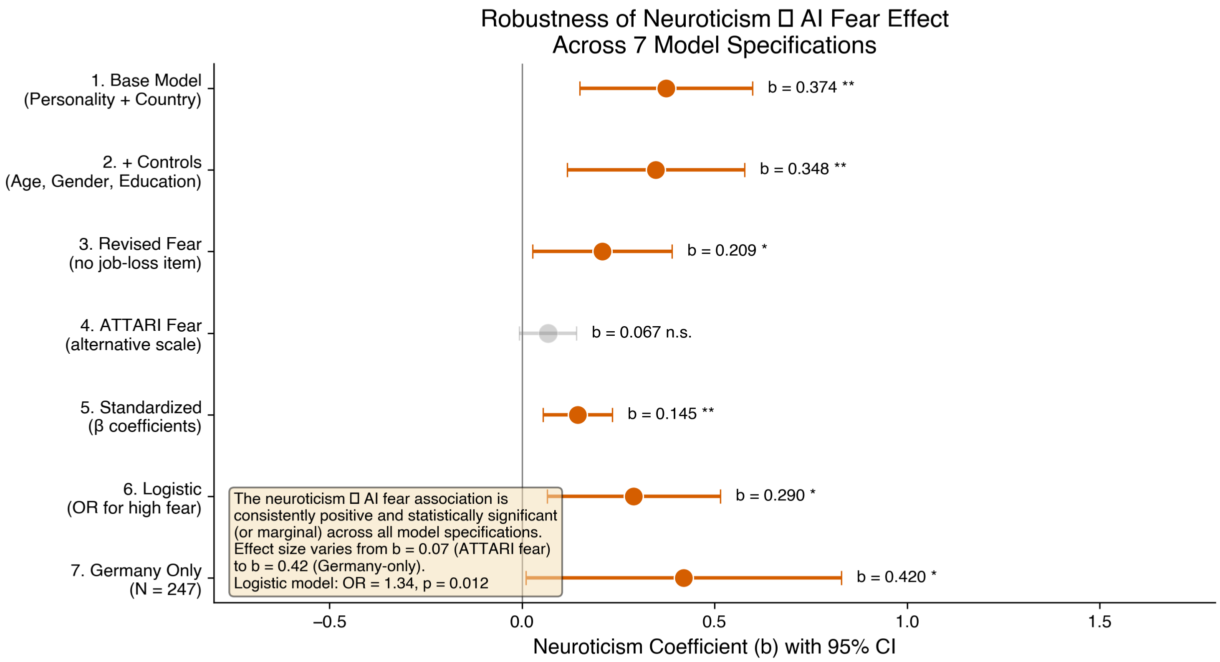

Figure 5: Robustness comparison showing the neuroticism → AI fear coefficient (b with 95% CI) across 7 model specifications. The effect is consistently positive across all specifications, reaching statistical significance in 6 of 7 models.

C. J. Soto and O. P. John. The next big five inventory (bfi-2): Developing and assessing a hierarchical model with 15 facets to enhance bandwidth, fidelity, and predictive power. *Journal of Personality and Social Psychology*, 113(1):117–143, 2017.

J.-P. Stein, B. Lieberoth, K. K. Andersen, and L. B. Andersen. Personality and attitudes toward artificial intelligence: A replication in a german sample. *Scientific Reports*, 14:5335, 2024.

G. B. Svendsen, J.-A. K. Johnsen, L. Almås-Sørensen, and J. Vittersø. Personality and technology acceptance: The influence of personality factors on the core constructs of the technology acceptance model. *Behaviour & Information Technology*, 32(5):505–516, 2011.

V. Taras, B. L. Kirkman, and P. Steel. Examining the impact of culture's consequences: A three-decade, multilevel, meta-analytic review of hofstede's cultural value dimensions. *Journal of Applied Psychology*, 95(3):405–439, 2010.

H. C. Triandis and M. J. Gelfand. Converging measurement of horizontal and vertical individualism and collectivism. *Journal of Personality and Social Psychology*, 74(1):118–128, 1998.

V. Venkatesh, M. G. Morris, G. B. Davis, and F. D. Davis. User acceptance of information technology: Toward a unified view. *MIS Quarterly*, 27(3):425–478, 2003.

M. Warr. Fear of crime in the united states: Avenues for research and policy. In D. Duffee, editor, *The Criminal Justice System*, pages 451–489. Greenwood, 2000.

A Full Model Diagnostics

Table 6 presents diagnostic statistics for the primary OLS models. All VIF values are below 1.2, indicating no multicollinearity concerns. Residual analysis reveals approximately normal errors (skewness = 0.471, kurtosis = 0.299) with no significant autocorrelation (Durbin-Watson = 1.648). Cook's distance identified 40 observations above the $4/N$ threshold; sensitivity analyses excluding these observations did not change the substantive conclusions.

B Correlation Matrix

Table 7 presents the Pearson correlation matrix among all continuous variables in the combined sample.

Table 6: Model Diagnostics

Diagnostic	Value	Threshold
Max VIF	1.197	<5.0
Mean VIF	1.115	<5.0
Residual skewness	0.471	≈0
Residual kurtosis	0.299	≈0
Durbin-Watson	1.648	≈2.0
High Cook's D (>4/N)	40	–
Cohen's f^2 (fear model)	0.131	small-to-medium
Cohen's f^2 (acceptance model)	0.120	small-to-medium

Table 7: Pearson Correlations Among Study Variables

Variable	1	2	3	4	5	6	7
1. Openness	–						
2. Conscientiousness	.11	–					
3. Extraversion	.18	.22	–				
4. Agreeableness	.05	.15	.07	–			
5. Neuroticism	−.12	−.20	−.26	−.07	–		
6. AI Acceptance	−.06	.09	.08	.16	−.08	–	
7. AI Fear	.01	−.08	.02	−.06	.15	−.28	–

Note. N = 615. Correlations $> |.08|$ are significant at $p < 0.05$.